

- Description

This module manages a smart fitness tracker system, ensuring that:

1. The system continuously monitors the user's heart rate (in beats per minute, BPM).
2. If the heart rate falls below a predefined minimum threshold, a low heart rate alert is triggered.
3. If the heart rate exceeds a predefined maximum threshold, a high heart rate alert is issued.
4. If the heart rate is within the optimal range, the system indicates that the heart rate is normal.

This system helps ensure exercise safety and optimal performance by promptly alerting the user of abnormal heart rates.

- Specification

module SmartHeartRateMonitoring

var currentHeartRate, minHeartRate, maxHeartRate: real

process updateHeartRate(newHeartRate: real) returns (currentHeartRate: real)

functional scenario form:

currentHeartRate < minHeartRate && output "ALERT: Heart rate is too low (" + currentHeartRate + " BPM).", Low heart rate alert triggered.

||

currentHeartRate > maxHeartRate && output "ALERT: Heart rate is too high (" + currentHeartRate + " BPM).", High heart rate alert triggered.

||

currentHeartRate >= minHeartRate && output "Heart rate is optimal (" + currentHeartRate + " BPM).", Heart rate is within the optimal range.

end process

end module

- Java Code

```
package code;
public class c84_SmartWaterPurifier {
    private double maxTDSThreshold;
    private double minTDSThreshold;
    private double currentTDS;
    private boolean filterCleaningActive;
    public c84_SmartWaterPurifier(double minTDSThreshold, double maxTDS
Threshold, double initialTDS) {
        this.minTDSThreshold = minTDSThreshold;
        this.maxTDSThreshold = maxTDSThreshold;
        this.currentTDS = initialTDS;
        this.filterCleaningActive = false;
    }
    public void updateTDS(double newTDS) {
```

```

        currentTDS = newTDS;
        adjustFilterActivation();
        System.out.printf("Current TDS: %.2f ppm, Filter Cleaning: %s\n",
                           currentTDS, filterCleaningActive ? "ACTIVE" :
                           "INACTIVE");
        if (filterCleaningActive) {
            System.out.println("ALERT: TDS level high! Initiating filter cleaning cycle.");
        } else {
            System.out.println("TDS level is within acceptable limits. No action required.");
        }
    }

    private void adjustFilterActivation() {
        if (currentTDS > maxTDSThreshold) {
            filterCleaningActive = true;
        } else if (currentTDS < minTDSThreshold) {
            filterCleaningActive = false;
        } else {
            filterCleaningActive = false;
        }
    }

    public double getCurrentTDS() {
        return currentTDS;
    }

    public boolean isFilterCleaningActive() {
        return filterCleaningActive;
    }

    public static void main(String[] args) {
        c84_SmartWaterPurifier purifier = new c84_SmartWaterPurifier(200.0, 400.0, 300.0);

        // testcase-VT:
        //     purifier.updateTDS(350.0);
        //     purifier.updateTDS(450.0);
        //     purifier.updateTDS(180.0);

        // testcase-FT:
        //     purifier.updateTDS(1.7976931348623157E308);
        //     purifier.updateTDS(-1.7976931348623157E308);
        //     purifier.updateTDS(0.0);

        // testcase-Z3:
        //     purifier.updateTDS(350.0);

```

```
//      purifier.updateTDS(450.0);
//      purifier.updateTDS(180.0);

// testcase-UVT:
purifier.updateTDS(350.0);
purifier.updateTDS(450.0);
purifier.updateTDS(180.0);
}
}
```

- Coverage report

1. Test case number: 3
2. V-Method testing branch coverage: 100%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
✓ VibrationTestingTool	0.5 %	4	838	842
✓ src	0.5 %	4	838	842
✓ code	0.5 %	4	838	842
✓ c85_SmartHeartRateMonitor.java	100.0 %	4	0	4
✓ c85_SmartHeartRateMonitor	100.0 %	4	0	4
main(String[])		0	0	0
c85_SmartHeartRateMonitor(do		0	0	0
updateHeartRate(double)	100.0 %	4	0	4

3. Vibration testing instruction coverage: 95.8%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
✓ VibrationTestingTool	0.3 %	68	21,155	21,223
✓ src	0.3 %	68	21,155	21,223
✓ code	0.3 %	68	21,155	21,223
✓ c85_SmartHeartRateMonitor.java	95.8 %	68	3	71
✓ c85_SmartHeartRateMonitor	95.8 %	68	3	71
main(String[])	100.0 %	17	0	17
c85_SmartHeartRateMonitor(do	100.0 %	12	0	12
updateHeartRate(double)	100.0 %	39	0	39

4. Fuzz testing branch coverage: 75%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
✓ VibrationTestingTool	0.4 %	3	839	842
✓ src	0.4 %	3	839	842
✓ code	0.4 %	3	839	842
✓ c85_SmartHeartRateMonitor.java	75.0 %	3	1	4
✓ c85_SmartHeartRateMonitor	75.0 %	3	1	4
main(String[])		0	0	0
c85_SmartHeartRateMonitor(do		0	0	0
updateHeartRate(double)	75.0 %	3	1	4

5. Fuzz testing instruction coverage: 91.5%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
✓ VibrationTestingTool	0.3 %	65	21,158	21,223
✓ src	0.3 %	65	21,158	21,223
✓ code	0.3 %	65	21,158	21,223
✓ c85_SmartHeartRateMonitor.java	91.5 %	65	6	71
✓ c85_SmartHeartRateMonitor	91.5 %	65	6	71
main(String[])	100.0 %	17	0	17
c85_SmartHeartRateMonitor(do	100.0 %	12	0	12
updateHeartRate(double)	92.3 %	36	3	39

6. Z3 branch coverage: 100%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
▼ VibrationTestingTool	0.5 %	4	838	842
▼ src	0.5 %	4	838	842
▼ code	0.5 %	4	838	842
▼ c85_SmartHeartRateMonitor.java	100.0 %	4	0	4
▼ c85_SmartHeartRateMonitor	100.0 %	4	0	4
main(String[])	100.0 %	0	0	0
c85_SmartHeartRateMonitor(do	100.0 %	0	0	0
updateHeartRate(double)	100.0 %	4	0	4

7. Z3 instruction coverage: 95.8%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
▼ VibrationTestingTool	0.3 %	68	21,155	21,223
▼ src	0.3 %	68	21,155	21,223
▼ code	0.3 %	68	21,155	21,223
▼ c85_SmartHeartRateMonitor.java	95.8 %	68	3	71
▼ c85_SmartHeartRateMonitor	95.8 %	68	3	71
main(String[])	100.0 %	17	0	17
c85_SmartHeartRateMonitor(do	100.0 %	12	0	12
updateHeartRate(double)	100.0 %	39	0	39

8. Unimproved V-method branch coverage: 100%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
▼ VibrationTestingTool	0.5 %	4	838	842
▼ src	0.5 %	4	838	842
▼ code	0.5 %	4	838	842
▼ c85_SmartHeartRateMonitor.java	100.0 %	4	0	4
▼ c85_SmartHeartRateMonitor	100.0 %	4	0	4
main(String[])	100.0 %	0	0	0
c85_SmartHeartRateMonitor(do	100.0 %	0	0	0
updateHeartRate(double)	100.0 %	4	0	4

9. Unimproved V-method instruction coverage: 95.8%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
▼ VibrationTestingTool	0.3 %	68	21,155	21,223
▼ src	0.3 %	68	21,155	21,223
▼ code	0.3 %	68	21,155	21,223
▼ c85_SmartHeartRateMonitor.java	95.8 %	68	3	71
▼ c85_SmartHeartRateMonitor	95.8 %	68	3	71
main(String[])	100.0 %	17	0	17
c85_SmartHeartRateMonitor(do	100.0 %	12	0	12
updateHeartRate(double)	100.0 %	39	0	39